



FARMERS MARKET
fresh and friendly

FINAL PROJECT - OOP

FARMERS MARKET

“PLANTING DREAMS, GROWING MEMORIES”

START

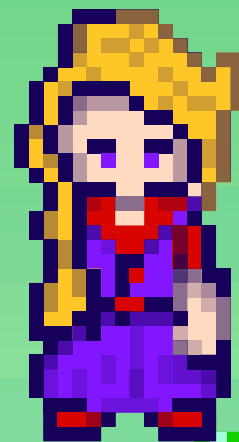
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INTRODUCTION



DESKRIPSI GAME

GAME SIMULASI FARMING 2D DENGAN FITUR GRID-BASED PLANTING, JUAL-BELI, DAN TOOLS PROGRESSIF DAN GAME INI MENGUJI STRATEGI DARI PEMAIN.



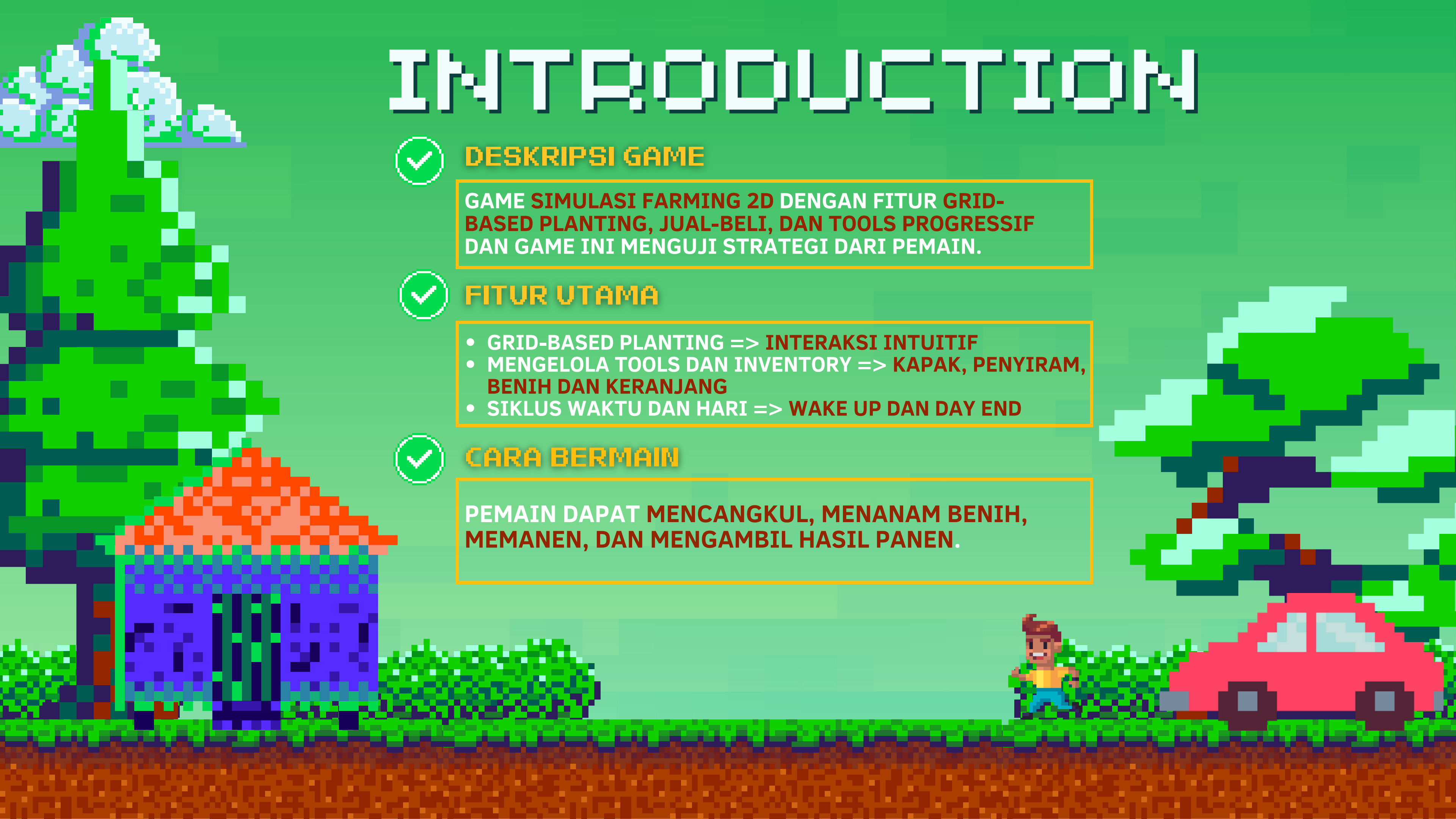
FITUR UTAMA

- GRID-BASED PLANTING => INTERAKSI INTUITIF
- MENGELOLA TOOLS DAN INVENTORY => KAPAK, PENYIRAM, BENIH DAN KERANJANG
- SIKLUS WAKTU DAN HARI => WAKE UP DAN DAY END



CARA BERMAIN

PEMAIN DAPAT MENCANGKUL, MENANAM BENIH, MEMANEN, DAN MENGAMBIL HASIL PANEN.



1

SCENE



1. SCENE MAIN



2. SCENE INDOORS

7AM



Kapak

Pengiram

Benih

Keranjang

3. DAY END

A New Day Begins
- Day X -

Press Any Button To
Continue

2

SCRIPT

CONTROLLER



1. PLAYER CONTROLLER

- MENGONTROL PERGERAKAN PEMAIN, PENGGUNAAN ALAT, DAN ANIMASI.

```
public class PlayerController : MonoBehaviour
{
    2 references
    public static PlayerController instance;

    0 references
    private void Awake()
    {
        if (instance == null)
        {
```

Memastikan hanya satu instance dari PlayerController (Singleton Pattern)

```
public float moveSpeed;

1 reference | 1 reference
public InputActionReference moveInput, actionInput;
```

Menghubungkan component

```
10 references
public enum ToolType
{
    3 references
    plough,
```

Mengatur visibilitas agar tidak bisa diakses sembarangan (Enum)

```
if(Keyboard.current.digit1Key.wasPressedThisFrame)
{
    currentTool = ToolType.plough;
    hasSwitchedTool = true;
```

Memilih tools "Plough" dengan digit "1"

```
void UseTool()
{
    GrowBlock block = null;
    //block = FindFirstObjectByType<GrowBlock>();
    //block.PloughSoil();

    block = GridController.instance.GetBlock(tool

    toolWaitCounter = toolWaitTime;

    if(block != null)
    {
        switch(currentTool)
        {
            case ToolType.plough:

                block.PloughSoil();
```

Method saat menggunakan tools yang terpilih dan (block)

2. GRID CONTROLLER

- PENANAMAN TANAMAN (POSISI DAN UKURAN GRID)

```
1 reference
public class GridController : MonoBehaviour
{
    1 reference
    public static GridController instance;
```

Memastikan hanya satu instance dari
PlayerController (Singleton Pattern)

```
private Vector2Int gridSize;
```

```
3 references
```

```
public List<BlockRow> blockRows = new List<BlockRow>()
```

Menyimpan elemen dalam "List<>"

```
0 references
```

```
void GenerateGrid()
```

```
{
    minPoint.position = new Vector3(Mathf.Round(minPoint.position.x), Mathf.Round(minPoint.position.y), Mathf.Round(minPoint.position.z));
    maxPoint.position = new Vector3(Mathf.Round(maxPoint.position.x), Mathf.Round(maxPoint.position.y), Mathf.Round(maxPoint.position.z));

    Vector3 startpoint = minPoint.position + new Vector3(.5f, .5f, 0f);
```

Mengatur grid dengan "Mathf"

Method untuk mengatur Growblock
dengan "Mathf"

```
public GrowBlock GetBlock(float x, float y)
{
    x = Mathf.RoundToInt(x);
    y = Mathf.RoundToInt(y);

    x -= minPoint.position.x;
    y -= minPoint.position.y;

    int intX = Mathf.RoundToInt(x);
    int intY = Mathf.RoundToInt(y);

    if(intX < gridSize.x && intY < gridSize.y)
    {
        return blockRows[intY].blocks[intX];
    }
}
```

Memastikan koordinat dalam
ukuran grid dan mengembalikan
blok yang sesuai atau null jika di
luar grid (Encapsulation)

3. CROP CONTROLLER

- INFORMASI DAN OPERASI TANAMAN
(PENAMBAHAN BENIH ATAU HASIL PANEN)

```
public class CropController : MonoBehaviour
{
    2 references
    public static CropController instance;

    0 references
    private void Awake()
    {
        if(instance == null)
        {
```

Memastikan hanya satu instance dari CropController (Singleton Pattern)

```
public enum CropType
{
    0 references
    pumpkin,
    0 references
    lettuce,
```

Mengatur visibilitas agar tipe tanaman tidak bisa diakses sembarangan (Enum)

```
0 references
public CropInfo GetCropInfo(CropType cropToGet)
{
    int position = -1;

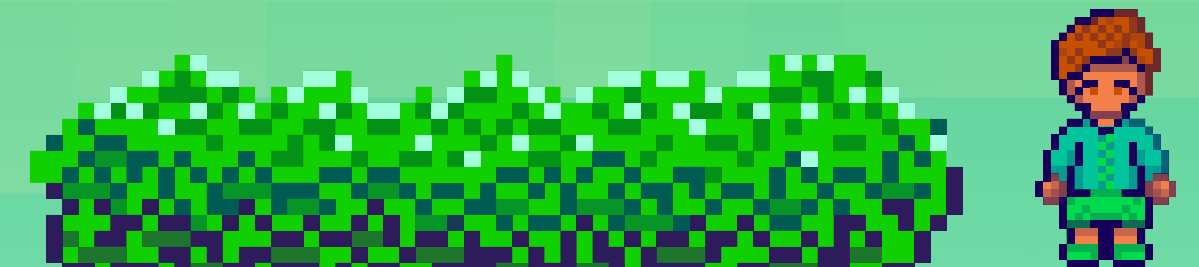
    for(int i = 0; i < cropList.Count; i++)
    {
        if (cropList[i].cropType == cropToGet)
        {
            position = i;
        }
    }
}
```

Menyimpan informasi lengkap tentang jenis tanaman, dikemas dalam satu class (Encapsulation)

Method untuk penambahan hasil panen atau benih

```
public void AddCrop(CropType cropToAdd)
{
    foreach (CropInfo info in cropList)
    {
        if (info.cropType == cropToAdd)
        {
            info.cropAmount++;
        }
    }
}

0 references
public void AddSeed(CropType seedToAdd, int amount)
{
    foreach(CropInfo info in cropList)
    {
        if(info.cropType == seedToAdd)
```



4. INVENTORY CONTROLLER

- MENGELOLA SISTEM INVENTARIS
(Menampilan dan memperbarui)

```
0 references
public class InventoryController : MonoBehaviour
{
    1 reference
    public SeedDisplay[] seeds;
    1 reference
    public CropDisplay[] crops;

    0 references
    public void OpenClose()
    {
        if (UIController.instance.theShop.gameObject.activeSelf == false)
        {
            if (gameObject.activeSelf == false)
            {
                gameObject.SetActive(true);
                UpdateDisplay();
            }
            else
            {
                gameObject.SetActive(false);
            }
        }
    }
}
```

Membuka/menutup inventaris untuk menyederhanakan manipulasi UI (Encapsulation)

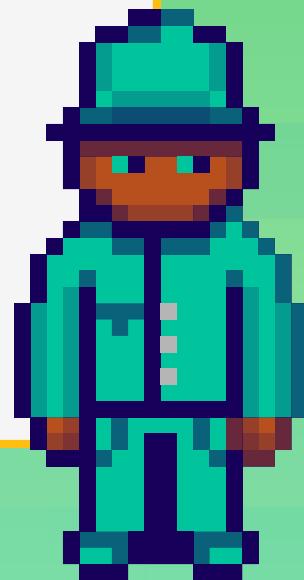
```
1 reference
public void UpdateDisplay()
{
    foreach(SeedDisplay seed in seeds)
    {
        seed.UpdateDisplay();
    }

    foreach(CropDisplay crop in crops)
    {
        crop.UpdateDisplay();
    }
}
```

Memanggil fungsi pada objek seeds dan crops meskipun tipe objek berbeda (Polymorphism)

Objek

Objek



5. CURRENCY CONTROLLER

- MENGELOLA SISTEM UANG PEMAIN
(**PENGELUARAN DAN PENAMBAHAN**)

```
6 references
public float currentMoney;

0 references
private void Start()
{
    UIController.instance.UpdateMoneyText(currentMoney);
}
```

Menampilkan nominal saat ini
pada UI Text

```
0 references
public void SpendMoney(float amountToSpend)
{
    currentMoney -= amountToSpend;
    UIController.instance.UpdateMoneyText(currentMoney);
}
```

Mengeluarkan uang dan
menampilkannya pada UI Text
(**Encapsulation**)

```
0 references
public void AddMoney(float amountToAdd)
{
    currentMoney += amountToAdd;
    UIController.instance.UpdateMoneyText(currentMoney);
}
```

Menambahkan uang dan
menampilkannya pada UI Text

Mengecek nominal uang apakah cukup
dengan bool (**TRUE/FALSE**)

```
public bool CheckMoney(float amount)
{
    if(currentMoney >= amount)
    {
        return true;
    } else
    {
        return false;
    }
}
```


E. TIME CONTROLLER

- MENGELOLA SISTEM WAKTU PEMAIN
(**END DAY DAN START DAY**)

```
1 reference
public void EndDay()
{
    timeActive = false;

    currentDay++;

    GridInfo.instance.GrowCrop();

    PlayerPrefs.SetString("Transition", "Wake Up");

    //StartDay();
    SceneManager.LoadScene(dayEndScene);
}
```

Merupakan logika malam hari dengan menonaktifkan waktu, menambah hari, dan memuat scene baru (**Abstraction**)

```
0 references
public void StartDay()
{
    timeActive = true;

    currentTime = dayStart;

    AudioManager.instance.PlaySFX(6);
}
```

Mengaktifkan siklus waktu (**Encapsulation**)

Mengatur ulang waktu ke awal hari

Memastikan hanya ada satu instance global yang mengatur efek suara (**Singleton**)

E. BED CONTROLLER

- MENGATUR PLAYER DALAM KEADAAN SEDANG
TIDUR ATAU TIDAK

```
private bool canSleep;

// Update is called once per frame
0 references
void Update()
{
    if(canSleep == true)
    {
        if(Mouse.current.leftButton.wasPressedThisFrame)
        {
            //GridInfo.instance.GrowCrop();

            if(TimeController.instance != null)
            {
                TimeController.instance.EndDay();
            }
        }
    }
}
```

Memeriksa player sedang tidur atau tidak

Objek yang memiliki tag "Player"
kemungkinan tidur (**canSleep = TRUE**)

```
private void OnTriggerEnter2D(Collider2D collision)
{
    if(collision.tag == "Player")
    {
        canSleep = true;
    }
}
```

Objek yang memiliki tag "Player"
kemungkinan tidak tidur (**canSleep = FALSE**)

```
private void OnTriggerExit2D(Collider2D collision)
{
    if(collision.tag == "Player")
    {
        canSleep = false;
    }
}
```



7. UI CONTROLLER

- **MENGATUR UI DARI FARMERS MARKET GAME (TOOLBAR, WAKTU, UANG, DAN JEDA)**

```
public static UIController instance;  
0 references  
private void Awake()  
{  
    if (instance == null)
```

→ Mengakses method dan variabel UIController script lain

```
public GameObject[] toolbarActivatorIcons;  
5 references  
public TMP_Text timeText;  
1 reference  
public InventoryController theIC;
```

→ Properti dalam UI yang terhubung dalam game

```
public void PauseUnpause()  
{  
    if(pauseScreen.activeSelf == false)  
    {  
        pauseScreen.SetActive(true);  
  
        Time.timeScale = 0f;  
    }  
    else  
    {  
        pauseScreen.SetActive(false);  
  
        Time.timeScale = 1f;  
    }  
  
    AudioManager.instance.PlaySFXPitchAdjusted(5);  
}
```

→ Method untuk pause dalam permainan Farmers Market game

Menampilkan jam menggunakan Text di UI dengan "Mathf.FloorToInt"

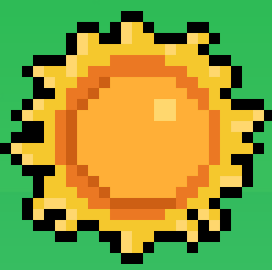
```
0 references  
public void UpdateTimeText(float currentTime)  
{  
    if(currentTime < 12)  
    {  
        timeText.text = Mathf.FloorToInt(currentTime) + "AM";  
    } else if(currentTime < 13)  
    {  
        timeText.text = "12PM";  
    } else if(currentTime < 24)  
    {  
        timeText.text = Mathf.FloorToInt(currentTime - 12) + "PM";  
    } else if(currentTime < 25)  
    {  
        timeText.text = "12AM";  
    } else  
    {  
        timeText.text = Mathf.FloorToInt(currentTime - 24) + "AM";  
    }  
}
```


3

AUDIO & ART

MANAGEMENT

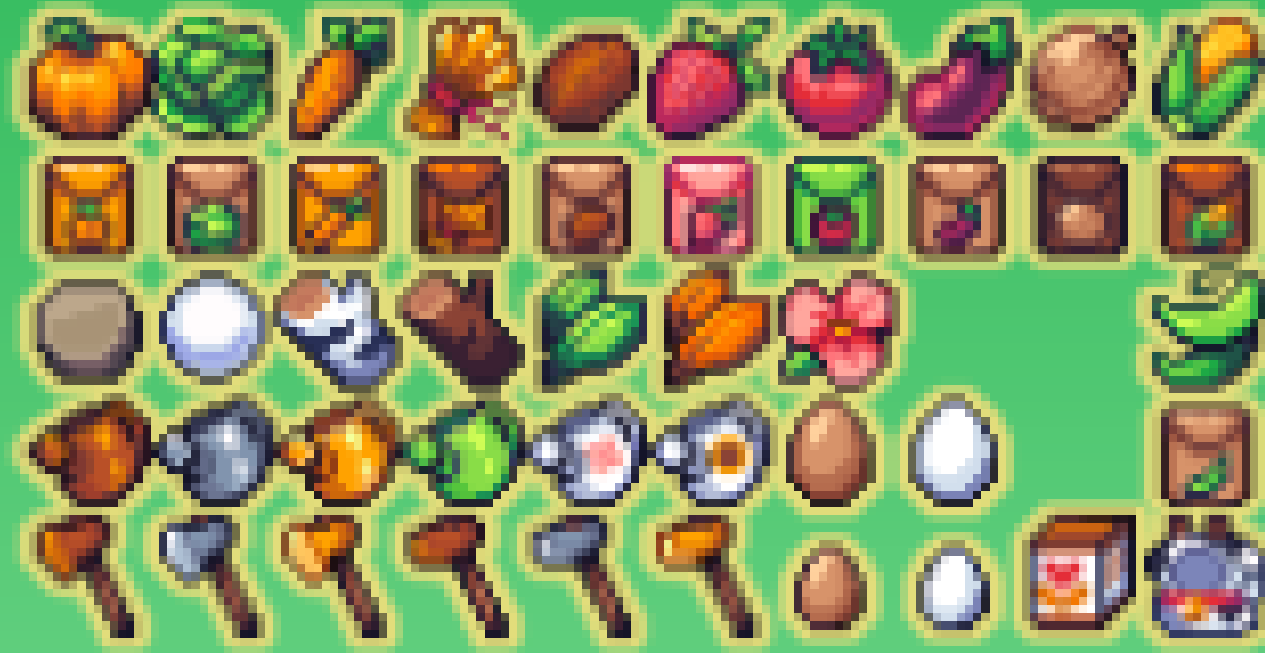




ART



FURNITURE



ITEMS



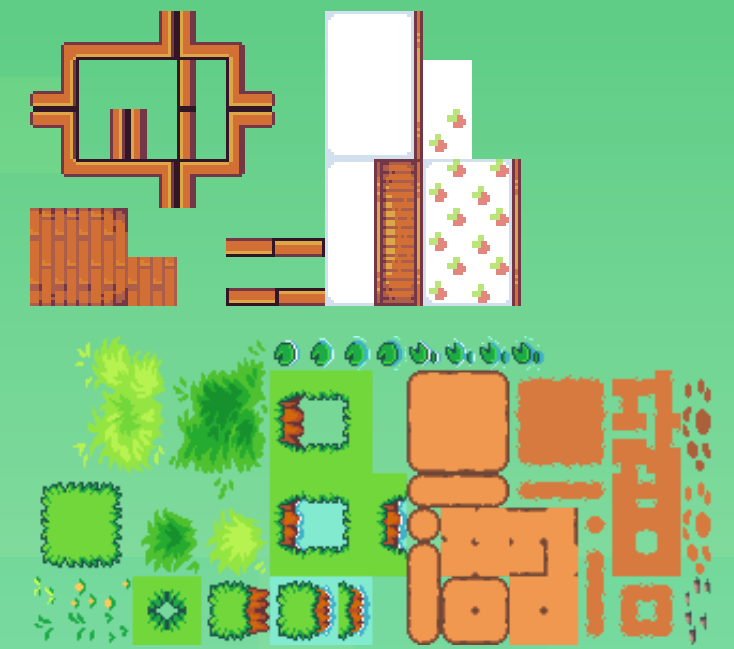
PLANTS



CHARACTER



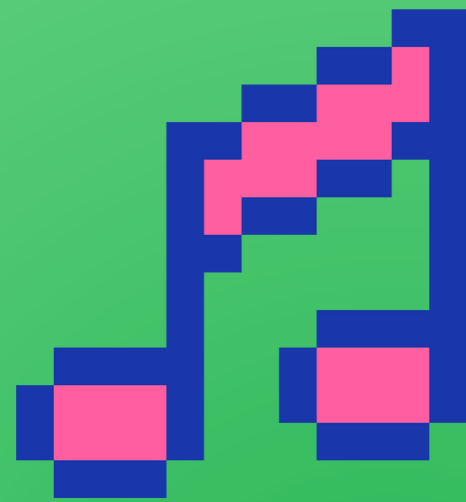
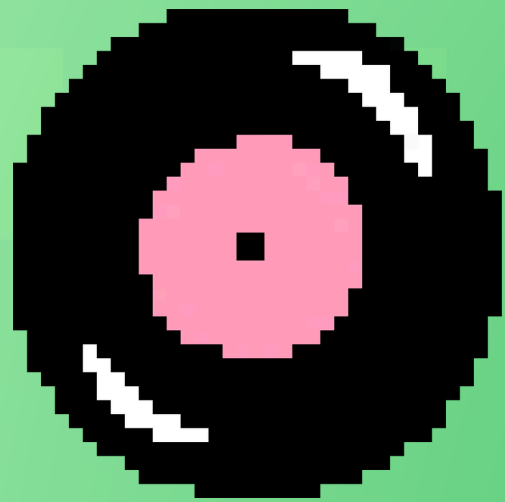
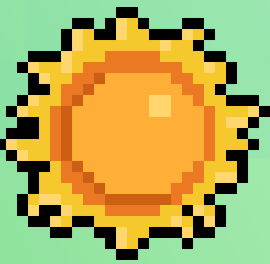
OBJECT



TILEMAP



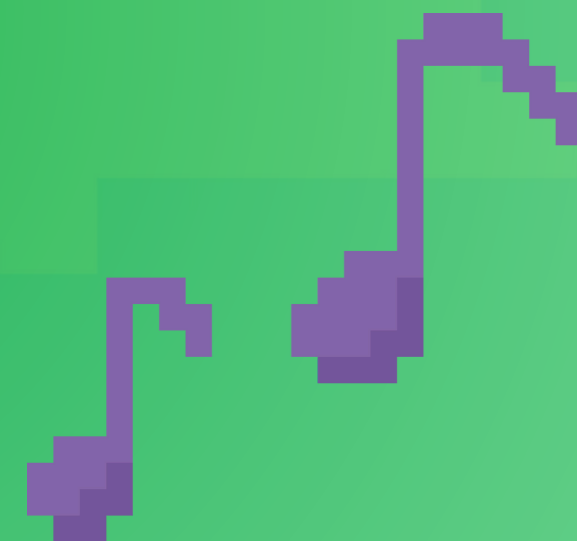
AUDIO



Apple Cider - Zane Little

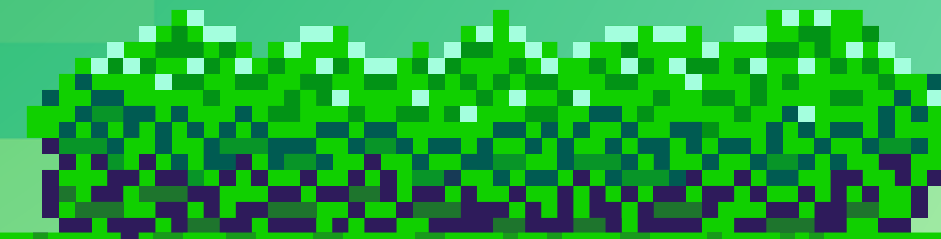
Hometown - Juhani Junkala

Bazaar - Juhani Junkala



Sound FX - James Doyle

Where Time Stands Still - Juhani Junkala





THANK
YOU